MCO-1D ORNITHOPTER

Complete Bill of Materials

Every component required to construct, wire, and commission the MCO-1D flapping-wing aircraft, down to individual fasteners and wire gauges.

Specifications derived from confirmed library constants and hardware commissioning documents.

All quantities for one complete aircraft unless noted. Revision 8 — Motor specification confirmed: GoolRC 5010-14, 12N14P, 7 pole pairs.

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SECTION 1 — AIRFRAME AND MECHANICAL STRUCTURE

All dimensions are confirmed from library constants. Wing semi-spans: W1=772mm, W2=749mm, W3=735mm, W4=712mm. Chord: 165mm uniform. Fore-aft stagger between pairs: 138mm. Vertical stagger (staircase): 100mm per step. Undercarriage height (wing root pivot to ground): 200mm.

#	Description	Part / Spec	Qty	Unit	Notes
1	**Carbon fibre spar — main (per wing)**	3mm OD × 772mm (W1 length)	16	pcs	2 spars per wing × 8 wings. Cut to length: W1=772, W2=749, W3=735, W4=712mm. Pultruded CF rod, minimum 40GPa stiffness.
2	**Carbon fibre spar — trailing edge**	1.5mm OD CF rod	8	pcs	1 per wing. Holds membrane taut along trailing edge. Cut to wing chord length.
3	**Wing membrane material**	Ripstop nylon, 40 denier, or Mylar film 12 micron	0.6	m²	Total wing area = 0.979 m². Add 20% for trimming. Ripstop preferred for durability. Mylar for lower weight.
4	**Fuselage spine**	CF tube 10mm OD × 4mm ID, total length ~700mm	1	pcs	Main structural member. All wing root joints, ODrives, and Pi mount to this. Length sized to fit 4 wing pairs at 138mm stagger spacing.
5	**Fuselage cross-members**	CF tube 6mm OD × 4mm ID	4	pcs	One per wing pair, perpendicular to spine. Approximately 200mm each. Carry wing root joints on each side.
6	**Sector wheels**	Custom machined or 3D printed, 200° arc, radius ~35mm	8	pcs	1 per wing. 200° total travel (±100° from centre). Must withstand rope tension forces. Aluminium preferred; PETG/ABS for prototyping. Groove for 1mm rope.
7	**Wing root pivot bearings**	MR84ZZ 4×8×3mm miniature ball bearing	16	pcs	2 per wing root joint (flap axis). Must be low-friction — any drag directly reduces efficiency.
8	**Sweep pivot bearings**	MR74ZZ 4×7×2.5mm or similar	8	pcs	1 per wing for sweep (forward/aft fold) pivot. Smaller than flap bearings.
9	**Drive rope**	Spectra/Dyneema 1mm, minimum 50kg breaking strength	4	m	Connects motor shaft capstan to sector wheel. 7:1 rope drive ratio. Spectra has near-zero stretch — essential for position control accuracy. Allow 0.5m per wing pair.
10	**Motor shaft capstans**	Turned aluminium, 8mm bore, ~10mm diameter	8	pcs	1 per motor. Rope wraps around this to drive the sector wheel. Must be secured to motor shaft with grub screw.

#	Description	Part / Spec	Qty	Unit	Notes
11	**Landing gear — main struts**	CF tube 6mm OD or aluminium wire 4mm bent to shape	2	pcs	Main wheels, one each side. Height to give 200mm from wing root pivot to ground. Lightweight — this is dead weight.
12	**Landing gear — tail strut**	CF rod 3mm OD, bent	1	pcs	Tail wheel support. Must prevent tail-strike during rotation.
13	**Wheels**	Lightweight foam or foam-filled rubber, 30-40mm diameter	3	pcs	2 main + 1 tail. Keep as light as possible. Model aircraft wheels suitable.
14	**Wing attachment hardware**	M2 × 8mm stainless steel screws and M2 nuts	40	sets	Securing wing hinges, servo mounts, and cross-members. Buy extra — easy to lose.
15	**Thread-lock adhesive**	Loctite 243 (medium strength)	1	bottle	Apply to all screws that must not vibrate loose during flight. Essential for motor mount screws.
16	**Epoxy adhesive**	5-minute epoxy, structural grade	1	pack	For bonding CF components and membrane to spars. West System or similar preferred.
17	**Dihedral control linkage rods**	M2 pushrod, 50mm, with ball-link ends	8	pcs	Connect sweep servo arm to wing root to control dihedral angle. Adjustable length preferred.

■ All carbon fibre work generates fine dust. Wear an N95 mask and work outdoors or with ventilation. CF dust is a lung irritant.

SECTION 2 — FLAP MOTORS

Confirmed specification from library: GoolRC 5010 360KV brushless outrunner motors. 8 required (one per wing). These are the primary thrust and lift producers — the equivalent of the engine on a conventional aircraft.

#	Description	Part / Spec	Qty	Unit	Notes
1	**Brushless outrunner motor**	GoolRC 5010 360KV	8	pcs	Confirmed hardware. 360KV = 360 RPM/V. At 14.8V nominal: max 5,328 RPM. With 7:1 rope drive gives adequate sector wheel speed for 3Hz flapping. Stator diameter: 50mm. Stack: 10mm.
2	**Motor mount plates**	Aluminium 2mm, custom cut to fit fuselage cross-members	8	pcs	Secure each motor to the fuselage cross-member. Must be rigid — any flex corrupts position control. Design to match motor bolt pattern.
3	**Motor mount screws**	M3 × 8mm stainless, with spring washers	32	pcs	4 screws per motor. Apply Loctite 243. Check tightness before every flight.
4	**Motor phase wires (extension)**	18AWG silicone wire, 3-conductor, red/yellow/black	2	m	If motor phase wires are too short to reach ODrive, extend. Keep as short as practical — resistance matters.
5	**Motor phase connectors**	3.5mm gold bullet connectors, male and female pairs	24	pairs	3 per motor × 8 motors = 24 pairs. For connecting motor phase wires to ODrive output. Solder carefully — poor joints cause overheating.
6	**Hall effect sensor cable**	6-pin JST-PH, 150mm	8	pcs	GoolRC 5010 has hall sensors for commutation. The connector from the motor must reach the ODrive encoder input. Extension cable if needed.

■ The GoolRC 5010 has no built-in winding temperature sensor. Motor thermal management relies on ODrive FET temperature monitoring and visual inspection. Allow motors to cool between extended test runs.

SECTION 3 — MOTOR CONTROLLERS (ODRIVES)

4 ODrive units required, each controlling 2 motors (one per channel). All communicate via CAN-FD bus. Critical setting: vel_limit = 150 rad/s on every axis.

#	Description	Part / Spec	Qty	Unit	Notes
1	**ODrive motor controller**	ODrive v3.6 (56V version) or ODrive S1	4	pcs	Each unit has 2 motor channels. 4 units × 2 channels = 8 motors total. Use 56V version for thermal headroom even though we run at 14.8V. CAN-FD bus communication.
2	**ODrive power cable**	XT30 male to bare wire, 16AWG, 150mm	4	pcs	Power input to each ODrive from PDB. 16AWG handles up to 20A continuous per ODrive.
3	**ODrive CAN bus cable**	Twisted pair 24AWG, CAT5 ethernet suitable	1	m	Daisy-chains all 4 ODrives and the Pi CAN interface. Cut from a short ethernet cable for pre-twisted pairs.
4	**CAN termination resistors**	120 ohm, 1/4W, ±1%	2	pcs	One at each END of the CAN chain. Pi end and last ODrive end. Without these the bus will not communicate reliably. Do not skip.
5	**ODrive USB cable (for configuration)**	USB-A to USB-C or Micro-USB (check ODrive model)	1	pcs	Needed during initial setup only. Connect ODrive to laptop to run ODrive Tool configuration. Not needed during flight.
6	**Heat sink compound**	Arctic Silver 5 or similar	1	tube	Apply between ODrive PCB and any heat sinking structure. Improves thermal dissipation during sustained operation.
7	**ODrive mounting standoffs**	M3 × 10mm brass standoffs	16	pcs	4 per ODrive for secure PCB mounting. Provides airflow underneath the board.

■ ODrive vel_limit MUST be set to 150 rad/s per axis before any flight testing. Factory default is much lower and will prevent the wings from reaching operating frequency. See Section 8 of the Commissioning Manual.

SECTION 4 — PITCH SERVOS

Confirmed specification: Helidirect TORQ TCM-1001. 8 required. Operate at 7.4V (from dedicated 2S LiPo or BEC), 333Hz PWM signal from Raspberry Pi GPIO. Speed: 1034°/s at 7.4V. Torque: 2.6 kg·cm at 7.4V. Running current per servo: approximately 180mA.

#	Description	Part / Spec	Qty	Unit	Notes
1	**Pitch servo**	Helidirect TORQ TCM-1001 digital servo	8	pcs	One per wing. Controls angle of attack (AoA) of each wing. Must be fast: ≥1200°/s at operating voltage. TCM-1001 provides 1034°/s at 7.4V — confirmed adequate. DO NOT SUBSTITUTE without recalculating.
2	**Servo power BEC (7.4V)**	2S LiPo 7.4V, 500mAh OR DC-DC BEC 14.8V→7.4V, 3A	1	pcs	8 servos × 180mA = 1.44A typical. Add 100% margin: 3A BEC minimum. Dedicated 2S LiPo (500mAh) provides clean power with no switching noise. If using BEC, use LC filter.
3	**Pitch servo extension leads**	JR/Futaba servo extension, 150mm, 3-wire	8	pcs	Extends servo lead from wing to fuselage wire run. Keep short to minimise weight.
4	**Servo mounting screws**	M2 × 6mm self-tapping, with servo grommets	32	pcs	4 per servo. Always use servo grommets (rubber mounts) to isolate vibration from servo body.
5	**Servo horns / arms**	Single-arm servo horn, compatible with TCM-1001 spline	8	pcs	One per servo. Check spline count matches TCM-1001 before ordering. Metal preferred for longevity.
6	**Servo linkage ball ends**	M2 ball-link, push-fit	16	pcs	2 per wing servo (one each end of the AoA control rod). Allows angular freedom at both ends.
7	**AoA control rods**	M2 threaded steel rod, 40mm	8	pcs	One per wing. Connects servo horn to wing root pitch pivot. Adjustable thread ends for neutral angle setting.
8	**Pitch servo PWM filter capacitors**	100µF electrolytic, 16V	8	pcs	Across VCC/GND of each servo connector at the GPIO header. Prevents PWM switching noise from corrupting Pi GPIO voltage.

■ The 333Hz PWM frequency for pitch servos is non-standard (most servos use 50Hz). Verify the TCM-1001 specification confirms 333Hz compatibility before use. The library is configured for this frequency and it cannot be changed without recalibrating the servo response.

SECTION 5 — SWEEP SERVOS

8 required. DS3225-class (25kg metal gear digital servo). Operate at 50Hz PWM, 5-6V supply. Sweep range: 0° (perpendicular) to 210° (fully stowed). Library SWEEP_LIMIT_MAX_DEG = 210° at full DS3225 range.

#	Description	Part / Spec	Qty	Unit	Notes
1	**Sweep servo**	DS3225 or equivalent 25kg metal gear digital servo	8	pcs	One per wing. Controls sweep angle (forward/aft fold of wing). 25kg·cm torque needed for wing structural loads during flight. Metal gear essential — plastic gears strip under load.
2	**Sweep servo power BEC (5-6V)**	DC-DC BEC 14.8V→5V or 6V, minimum 4A output	1	pcs	8 servos × max 500mA stall = 4A peak. 5V for standard DS3225, 6V for higher torque if supported. Check servo datasheet for max voltage.
3	**Sweep servo extension leads**	JR/Futaba servo extension, 200mm, 3-wire	8	pcs	Same as pitch servos but slightly longer as sweep servos are closer to wing roots.
4	**Sweep servo mounts**	Custom bracket, aluminium 2mm or 3D printed nylon	8	pcs	Secure sweep servo to fuselage cross-member. Must prevent servo rotation under load.
5	**Sweep servo horns**	Double-arm or wheel horn, DS3225 spline compatible	8	pcs	Wheel horn preferred — provides symmetric force on both sweep directions. Check spline size.
6	**Sweep linkage rods**	M3 threaded rod, 60mm, with M3 ball ends	8	pcs	Connects servo to wing sweep pivot. M3 for sweep (larger than pitch) due to higher forces.

SECTION 6 — RASPBERRY PI FLIGHT COMPUTER

The Raspberry Pi 4 is the main flight computer running the Wave Library. GPIO pins provide PWM to 16 servos. CAN interface connects to 4 ODrives. Serial UART connects to the Pixhawk.

#	Description	Part / Spec	Qty	Unit	Notes
1	**DS18B20 OAT Sensor**	Waterproof variant, One-Wire, 0.5 degree accuracy	1	pcs	Outside air temperature sensor for density altitude compensation. Connect to GPIO4 (One-Wire bus). No additional package required — standard Pi OS One-Wire kernel driver. Enable in /boot/config.txt: dtoverlay=w1-gpio,gpiopin=4 Read with: cat /sys/bus/w1/devices/28-*/w1_slave At 1500m with 5C OAT, stall speed corrects from 4.85 to 5.22 m/s automatically — critical for backcountry SAR at elevation. Approximately 3 USD. Mount in airflow away from motor heat sources.
1	**USB-A Flash Drive**	8 GB minimum, any brand	1	pcs	Required for knowledge transfer between aircraft. After each flight session the post-flight sequence automatically exports all discovered modes and validated principles to mco_knowledge_NNN.json. Plug the drive into the Pi USB port and run: knowledge_export_usb from the ground station. Drive then carries the knowledge file to any future MCO aircraft. Standard USB-A connection. Recommend keeping one permanently with the ground kit. No special format required — FAT32 or exFAT both work. CONFIRMED SPECIFICATION: 12N14P configuration — 12 stator slots, 14 magnetic poles, 7 pole pairs (14 poles divided by 2). ODrive setting: axis.motor.config.pole_pairs = 7. Wave library constant: MOTOR_POLE_PAIRS = 7. No field verification needed.

#	Description	Part / Spec	Qty	Unit	Notes
1	**Official Raspberry Pi 7-inch DSI Touchscreen**	Official Foundation product	1	pcs	Ground panel touchscreen. Connects via DSI ribbon connector and GPIO power — zero driver installation. Capacitive multi-touch, 800x480, 70mm x 130mm active area. Runs mco_panel.py (4 buttons: PRE-FLIGHT, POST-FLIGHT, STORM STOW, EMERGENCY STOP). Install with: sudo apt install python3-pygame
1	**Raspberry Pi 4 Model B**	4GB RAM version recommended	1	pcs	Main flight computer. 4GB RAM gives headroom for the 33,472-line Wave Library, 9 optimizer loops, 32-dimension autonomous discovery, 4-layer exploration system (biomimetic, evolutionary, random safety-gated), and knowledge transfer. Full software stack: 10 Python files, 42,319 lines total. 18 active biomimetic hypotheses spanning dragonfly, bat, manta, pterosaur, hawkmoth, locust, plesiosaur, sea turtle, sea lion, and otter. 72 hypothesis variants. 255 failure combinations. 4GB RAM recommended.
2	**MicroSD card — HIGH ENDURANCE type**	Samsung PRO Endurance 64GB or SanDisk High Endurance 64GB. NOT the standard EVO/Ultra variants.	1	pcs	CRITICAL SELECTION: The Wave Library logs flight data every 20ms — 50 small random writes per second continuously. A standard "Class 10" card sustains 10MB/s sequential but degrades to under 1MB/s for random small writes. This can cause tick overruns and flight control jitter. The PRO Endurance and High Endurance cards are specifically rated for dashcam/CCTV duty — identical write pattern to flight logging. They maintain >4MB/s random write performance indefinitely. Do not substitute a standard card.
3	**Pi official USB-C power supply**	5V 3A USB-C, official or quality equivalent	1	pcs	ONLY for ground testing. In flight use a 5V BEC from the main battery. The Pi is very sensitive to voltage drops — a bad supply causes random crashes and corrupted SD cards.

#	Description	Part / Spec	Qty	Unit	Notes
4	**5V BEC for flight**	DC-DC step-down, 14.8V→5V, minimum 3A, USB-C output or direct 5V rail	1	pcs	Powers Pi from main LiPo during flight. Must be a quality regulated supply. Holybro or Matek BECs are reliable.
5	**CAN hat for Raspberry Pi**	MCP2515 CAN Bus Shield (SPI interface)	1	pcs	Plugs onto Pi GPIO header. Provides CAN-FD interface to the 4 ODrives. SPI bus connection to Pi. Must enable SPI in raspi-config.
6	**Pi GPIO breakout board**	40-pin ribbon cable + breadboard breakout	1	pcs	Makes servo wiring to GPIO pins manageable. Route 16 servo signal wires from breakout to servo rail.
7	**Pi case / mounting plate**	Lightweight open-frame case or bare aluminium mounting plate	1	pcs	Protects Pi from vibration damage while allowing airflow for cooling. Full enclosures trap heat — use open frame or passive heatsink.
8	**Pi heatsink set**	Adhesive copper heatsinks for CPU, RAM, USB chip	1	set	The Pi 4 runs warm under load. Heatsinks prevent thermal throttling during long test sessions. Three-piece set covers all hot chips.
9	**GPIO ribbon cable**	40-pin rainbow ribbon, 150mm	1	pcs	Connects Pi GPIO to breakout board. Keep short to minimise interference.
10	**Serial-to-UART level shifter**	3.3V/5V bi-directional level converter, 4-channel	1	pcs	For Pi serial UART to Pixhawk if Pixhawk runs at 5V logic. The Pi GPIO is 3.3V. Check Pixhawk datasheet — many are 3.3V tolerant and do not need the shifter.

SECTION 7 — PIXHAWK AUTOPILOT

The Pixhawk provides the IMU (attitude), barometric altimeter, GPS, and runs ArduPlane firmware. Communicates with the Pi at 921600 baud via MAVLink on the TELEM2 port.

#	Description	Part / Spec	Qty	Unit	Notes
1	**Pixhawk autopilot**	Pixhawk 4 Mini, Pixhawk 6C, or Cube Orange (all compatible)	1	pcs	Any ArduPilot-compatible Pixhawk variant. Cube Orange preferred for triple-redundant IMU. Pixhawk 4 Mini is lightest option.
2	**GPS module**	Here3 or M8N GPS with compass, with mast	1	pcs	Mount on a mast above the fuselage for best sky view. Integrated compass needed for heading hold. Keep away from motor wiring (magnetic interference).
3	**Pitot tube (airspeed sensor)**	Differential pressure sensor, MS4525DO or equivalent	1	pcs	Measures airspeed. The Wave Library uses V_ms for back-side detection (6.3 m/s threshold). Pitot faces forward, away from wing downwash.
4	**Pixhawk power module**	PM07 or compatible, 3S-6S, 90A rated	1	pcs	Provides 5V to Pixhawk and measures battery voltage and current. Connects between battery and PDB.
5	**Telemetry radio (optional)**	SiK 915MHz telemetry pair, or RFD900	1	pair	For extended-range ground station. Not required for tethered TC-02/TC-03 tests where ethernet cable is used. Useful for free flight monitoring.
6	**RC receiver**	Compatible with your transmitter (SBUS, CRSF, or PPM output)	1	pcs	Receives pilot stick and lever commands. Connect to Pixhawk RC input. Use SBUS or CRSF for best latency.
7	**RC transmitter**	Minimum 6-channel, FrSky, Radiomaster, or Spektrum	1	pcs	Pilot control device. Needs at least: Left stick (pitch/yaw), Right stick (roll/collective lever), lever slider, plus 2 switches for mode selection.
8	**TELEM2 serial cable**	6-pin JST-GH to 3-pin Dupont (TX, RX, GND)	1	pcs	Connects Pixhawk TELEM2 port to Pi serial UART. Pin 2=TX, Pin 3=RX, Pin 6=GND. TX and RX cross (Pixhawk TX → Pi RX, Pixhawk RX → Pi TX).

SECTION 8 — LOAD CELLS AND TEST EQUIPMENT

Required for TC-02 (static) and TC-03 (dynamic) thrust tests. The HX711 ADC reads the load cell. Pi GPIO pins: DT=GPIO26, SCK=GPIO7.

#	Description	Part / Spec	Qty	Unit	Notes
1	**Load cell (bar type)**	50N or 100N rated, 4-wire strain gauge	1	pcs	Measures upward thrust. Aircraft weight is 33.8N — use 50N minimum, 100N recommended for margin. Bar-type (S-beam or single-point) preferred for clean axial loading.
2	**HX711 ADC amplifier board**	HX711 24-bit ADC module	1	pcs	Amplifies and digitises load cell signal. Very common, inexpensive. RATE pin: set HIGH for 80Hz sample rate (needed for per-stroke averaging). Wires: DT→GPIO26, SCK→GPIO7, VCC→3.3V, GND→GND.
3	**Load cell calibration weight**	Known mass: 1kg and 2kg calibration weights	1	set	Needed to calibrate the load cell before TC-02. Place known weight, record reading, set scale factor. Must be accurately known mass.
4	**Test rig structure**	Steel or heavy timber frame, minimum 10kg mass	1	pcs	Must be rigid and heavy enough that vibrating wings do not move it. Weld or bolt — do not use wood screws alone. Secure to floor if possible.
5	**Tether attachment hardware**	M8 eyebolt + M8 nut + nylon lock nut	2	sets	Safety tether attachment points. Even during static testing, tether the aircraft in case a rig failure allows unexpected movement.
6	**Safety tether rope**	Dyneema 3mm, 500kg breaking, 2m length	1	pcs	Tethers aircraft to rig during TC-02. Long enough to allow full test but short enough to prevent impact if rig fails.
7	**Wing clearance guards**	Soft foam padding on rig uprights within wing arc	1	set	In case a wing strikes the rig structure during a test, foam prevents damage. The wing tips travel in a 96° arc at speed.
8	**Digital multimeter**	Any quality DMM with DC voltage and continuity	1	pcs	Essential for verifying wiring before power-on. Check all connections before first test.
9	**Bench power supply**	Variable DC, 0-20V, 5A minimum	1	pcs	For ODrive configuration and motor calibration (set to 14.8V, limit 2A). Safer than battery for bench work — current limiting prevents damage.

SECTION 9 — POWER SYSTEM

The main battery is a 4S LiPo at nominally 14.8V. Power distribution feeds: ODrives (direct), pitch servos (7.4V), sweep servos (5-6V), Raspberry Pi (5V), and Pixhawk (5V via PM).

#	Description	Part / Spec	Qty	Unit	Notes
1	**Main flight battery**	4S LiPo, 14.8V nominal, 4000-6000mAh, 45C or higher	1	pcs	Primary power source. 45C rating means it can deliver 45 × capacity in amps — a 5000mAh 45C delivers 225A peak (far more than needed). Tattu, Turnigy, or CNHL are reliable brands. NEVER use budget no-name cells.
2	**LiPo battery charger**	Balance charger, 4S compatible, minimum 50W (e.g. ISDT Q6 Plus)	1	pcs	MUST be a balance charger — charges each cell individually to prevent imbalance. Never charge LiPo with a basic DC power supply.
3	**LiPo safe charging bag**	LiPo fire-retardant bag, large enough for battery	1	pcs	Always charge inside this bag. If battery catches fire during charge the bag contains it. Not optional.
4	**Power distribution board (PDB)**	XT60 input, multiple XT30/XT60 outputs, integrated BECs preferred	1	pcs	Distributes battery power to all consumers. Look for a PDB with integrated 5V and 12V BEC outputs to reduce component count. Holybro or Matek options suitable.
5	**Main battery connector**	XT60 female (battery side) and XT60 male (PDB side)	1	pair	XT60 is the standard high-current connector for this power level. Solder with care — poor solder joints cause resistance heating and failure.
6	**Anti-spark connector**	XT60 with built-in resistor or separate XT90-S	1	pcs	Prevents the large spark when connecting the battery. Spark can damage ODrive capacitors and is a fire risk near LiPo cells.
7	**ODrive power cables**	XT30 male to XT30 female, 16AWG, 100mm	4	pcs	One per ODrive from PDB. 16AWG handles 20A continuous — more than one ODrive channel needs. Keep short to minimise resistance.
8	**5V BEC for Raspberry Pi**	DC-DC step-down, 14.8V→5V, 3A min, USB-C output preferred	1	pcs	Dedicated supply for Pi only. Do not share with servos — servo current spikes will crash the Pi. Holybro or Matek recommended.

#	Description	Part / Spec	Qty	Unit	Notes
9	**7.4V BEC for pitch servos**	DC-DC step-down 14.8V→7.4V, 3A minimum	1	pcs	8 pitch servos at 180mA each = 1.44A typical. 3A for margin. Or use a dedicated 2S LiPo (500mAh) for completely clean power.
10	**5-6V BEC for sweep servos**	DC-DC step-down 14.8V→5V, 4A minimum	1	pcs	8 sweep servos at 500mA stall each. Size for peak, not average. Keep separate from pitch servo supply for noise isolation.
11	**Battery voltage alarm**	LiPo cell voltage alarm, 2S-6S compatible	1	pcs	Plugs into balance connector and beeps loudly when any cell drops below a set threshold (set to 3.6V). Mandatory for outdoor flight.
12	**Wiring harness fuse**	ANL fuse 30A with holder, or blade fuse 30A in-line	1	pcs	In-line fuse between battery and PDB. Protects wiring in case of short circuit. Without this a short can cause a LiPo fire.

SECTION 10 — WIRING, CONNECTORS AND FASTENERS

#	Description	Part / Spec	Qty	Unit	Notes
1	**Silicone wire 18AWG**	Red and black, silicone insulation, multi-strand	3	m	Power runs: battery to PDB, PDB to ODrives. Silicone insulation is flexible and heat resistant — do not use PVC wire near ESCs.
2	**Silicone wire 22AWG**	Multiple colours	2	m	Signal wiring: servo extensions, sensor connections, CAN bus. Lighter gauge reduces weight.
3	**Silicone wire 24AWG**	Red, black, white, green for load cell	0.5	m	HX711 to Pi wiring. Short runs, low current.
4	**Heat shrink tubing**	Assorted sizes 1mm-8mm, pack of 200	1	pack	Over all soldered joints. Use correct size — too large does not shrink enough to seal.
5	**Cable ties (small)**	100mm × 2.5mm nylon, pack of 100	1	pack	Securing wire runs to fuselage. Keep all wires away from wing flap paths.
6	**Servo connector pins**	JR/Futaba female 3-pin servo connectors, pack of 20	1	pack	For custom-length servo leads or any connector that needs replacement.
7	**JST-PH connectors**	2-pin and 3-pin JST-PH male/female pairs	10	pairs	For small signal connections: hall sensor extensions, HX711 to Pi.
8	**Dupont connectors and pins**	2.54mm pitch female housing and pins, assorted	1	kit	For Pi GPIO connections. Crimp tool needed. Or use pre-made jumper wires.
9	**Pre-made jumper wires**	Female-to-female 200mm, pack of 40 in multiple colours	1	pack	Simpler alternative to crimped Dupont connectors for Pi GPIO connections. More reliable than hand-crimped for prototyping.
10	**Electrical tape**	Self-amalgamating rubber tape, 19mm wide	1	roll	Self-amalgamating (not standard PVC) for waterproofing any joint that may get wet.
11	**Solder**	60/40 rosin-core solder, 0.8mm diameter	1	reel	For all electrical connections. Lead-free is available but harder to work with — 60/40 gives better joints for beginners.
12	**Flux pen**	No-clean flux pen	1	pcs	Dramatically improves solder joints on motor phase connections and XT60 connectors. Clean flux type does not require washing after.

SECTION 11 — CAN BUS HARDWARE

The CAN-FD bus connects the Raspberry Pi to all four ODrives. All devices share a single pair of wires in a daisy-chain. Both ends of the chain MUST have 120Ω termination resistors.

#	Description	Part / Spec	Qty	Unit	Notes
1	**MCP2515 CAN hat for Raspberry Pi**	MCP2515 SPI CAN Bus module with TJA1050 transceiver	1	pcs	Plugs onto Pi 40-pin header. Provides CAN interface via SPI. Set CAN speed to 1 Mbit/s. Enable SPI in raspi-config first. Install: sudo apt install can-utils.
2	**CAN bus cable**	CAT5e ethernet cable or twisted pair 24AWG	0.5	m	Cut a short ethernet cable and use one twisted pair (e.g. orange/white-orange) for CAN-H and CAN-L. Pre-twisted pairs are essential for noise rejection.
3	**CAN termination resistors**	120Ω ±1%, 1/4W, through-hole	2	pcs	One at Pi end of CAN chain, one at last ODrive end. Solder across CAN-H and CAN-L at each termination point. Do not omit.
4	**CAN connector (board edge)**	2-pin screw terminal or JST connector for CAN-H/CAN-L	5	pcs	1 per ODrive + 1 for Pi hat. Screw terminals simplest for field maintenance.

■ To test the CAN bus before connecting hardware: sudo ip link set can0 up type can bitrate 1000000 then: cansend can0 001#DEADBEEF and: candump can0. If you see the message echoed, the bus is working.

SECTION 12 — LAPTOP AND GROUND STATION

#	Description	Part / Spec	Qty	Unit	Notes
1	**Laptop computer**	Any Mac, Linux, or Windows laptop with Python 3.8+	1	pcs	Runs the ground station client (mco_ground_station.py). No special performance needed — the application is lightweight. Screen size: larger is easier to read live data.
2	**USB-to-ethernet adapter**	USB-A or USB-C to gigabit ethernet (if laptop has no port)	1	pcs	Necessary if laptop has no built-in ethernet port. Any quality brand works. Avoid very cheap adapters — they can drop packets.
3	**Ethernet cable**	Cat5e or Cat6, 3m	1	pcs	Direct Pi-to-laptop connection. Any standard ethernet cable. No crossover needed — modern adapters auto-negotiate.
4	**Python 3.8 or later**	From python.org or system package manager	1	install	Required to run the ground station client. No additional Python packages needed — the ground station uses only the standard library. Revision 3 note: all new capabilities (HSF dynamic sweep, progressive stowing, autonomous discovery, Loop 9 meta-optimizer) are pure software additions. No new hardware is required beyond what is listed in this document.
5	**ODrive Tool (for configuration)**	pip install odrive	1	install	Needed once for ODrive configuration. Run on the laptop connected via USB to each ODrive. Not needed after initial setup.
6	**Mission Planner or QGroundControl**	Free download from ardupilot.org or qgroundcontrol.com	1	install	For Pixhawk configuration and firmware update. Windows only for Mission Planner; QGC is cross-platform. Only needed for initial Pixhawk setup.

SECTION 13 — CONSUMABLES, TOOLS AND SAFETY EQUIPMENT

#	Description	Part / Spec	Qty	Unit	Notes
1	**Soldering iron**	60W temperature-controlled, e.g. Hakko FX-888D or TS100 portable	1	pcs	Temperature control essential. Set 320-370°C for 60/40 solder. Chisel tip for XT60, fine tip for signal wiring.
2	**Wire strippers**	Automatic, 0.2-6mm range	1	pcs	For preparing all wire ends. Automatic type strips to exact depth without nicking strands.
3	**Crimping tool**	Dupont/JST crimping tool, ratchet type	1	pcs	For making custom servo lead and sensor connector ends. Ratchet type ensures consistent crimp force.
4	**Drill and drill bits**	Cordless drill + bits 2mm, 3mm, 4mm, 6mm	1	set	For motor mount holes, fuselage attachment points, and wire routing holes.
5	**Hex driver set**	Metric hex keys 1.5mm through 4mm	1	set	For M2, M3 grub screws and motor mounting. Ball-end type allows angle access.
6	**Digital calipers**	150mm digital vernier calipers	1	pcs	Measuring wing dimensions, checking fit of mechanical parts, verifying gear clearances.
7	**Scale/balance**	Kitchen scale, 5kg, 1g resolution	1	pcs	Weighing the completed aircraft for MTOW confirmation. MTOW confirmed at 3.448 kg.
8	**Safety glasses**	ANSI Z87 rated impact protection	2	pcs	One per person present during any motor or wing test. Carbon fibre splinters and rope failures are eye hazards.
9	**Ear protection**	30dB NRR earplugs or earmuffs	2	pcs	Motor + wing noise during TC-02 can exceed 85dB at 1m. Required for anyone within 3m of the rig.
10	**Fire extinguisher**	ABC dry powder, 1kg minimum, or Class D for metal fires	1	pcs	Keep within 3m of any LiPo charging or motor testing area. LiPo fires spread very fast. Know how to use it before testing.
11	**First aid kit**	Basic kit including burn dressings	1	pcs	For soldering burns and cuts from carbon fibre. Know location before starting work.
12	**Cable labels**	Self-adhesive labels or heat-shrink labels	1	pack	Label every wire at both ends during construction. A label now saves an hour of tracing later. Use: wing number, servo type (P=pitch, S=sweep), port/starboard.

#	Description	Part / Spec	Qty	Unit	Notes
13	**Isopropyl alcohol**	99% IPA, 250ml	1	bottle	Cleaning solder flux residue and PCB contacts before assembly. Do not use on servo gears or bearings.
14	**Thread-lock adhesive**	Loctite 243 medium (BLUE), 5ml	1	tube	Apply to all motor mount, servo mount, and landing gear screws. BLUE not RED — red requires heat to remove and may strip small screws.

SECTION 14 — COST SUMMARY

Approximate costs in USD at time of writing. Prices vary by supplier and region. Electronics prices in particular fluctuate. Budget an additional 20% for shipping, import duties, and replacement of any damaged items.

14.1 By Category

Airframe and mechanical (CF, bearings, rope, hardware)	~$120
Flap motors × 8 (GoolRC 5010 360KV)	~$160
ODrives × 4	~$480
Pitch servos × 8 (Helidirect TORQ TCM-1001)	~$240
Sweep servos × 8 (DS3225 class)	~$80
Raspberry Pi 4 + accessories	~$110
Pixhawk + GPS + airspeed	~$180
Load cell + HX711 + test rig hardware	~$60
Power system (batteries, BECs, PDB, charger)	~$120
Wiring, connectors, CAN hardware	~$40
Tools and consumables	~$120
Laptop (if not already owned)	~$0–800
[redacted]	**[redacted]**
SUBTOTAL (excluding laptop)	~$1,710
Contingency +20%	~$340
TOTAL ESTIMATE	~$2,050

■ The single largest variable cost is the ODrives. Budget options exist but are not recommended for this application — position control accuracy directly affects flight quality. The GoolRC 5010 motors and Helidirect TCM-1001 servos are confirmed hardware and should not be substituted without recalibrating the Wave Library.

14.2 Critical Items — Do Not Substitute

The following items have confirmed specifications embedded in the Wave Library. Substituting them without updating the library constants will result in incorrect flight behaviour:

GoolRC 5010 360KV motors	KV rating and pole pairs are used in CT calculations
Helidirect TORQ TCM-1001 pitch servos	Speed (1034°/s), torque (2.6 kg·cm), 333Hz operation all calibrated
4S LiPo main battery (14.8V nominal)	Voltage used in ODrive configuration and thermal monitoring

ODrive v3.6 or S1	CAN-FD protocol and parameter names differ between ODrive generations

End of MCO-1D Bill of Materials

Cross-reference with Commissioning Manual for installation details of each component.